36014

Haynes

# Ford Edge & Lincoln MKX

## 2007 thru 2014 □ All models

# Haynes Repair Manual

**Based on a complete teardown and rebuild**


PRINTED ON RECYCLED PAPER

**Includes essential information for today's more complex vehicles**

# 2008 TRANSMISSIONS

## Four Wheel Drive (4WD) Systems - Edge & MKX

## SPECIFICATIONS

**MATERIAL**

| Item | Specification | Fill Capacity |
|---|---|---|
| Motorcraft SAE 75W-140 Synthetic Rear Axle Lubricant XY-75W140-QL (US); CXY-75W140-1L (Canada) | WSL-M2C192-A | 530 ml (18 oz)[a] |
| Motorcraft SAE 80W-90 Premium Rear Axle Lubricant XY-80W90-QL (US); CXY-80W90-1L (Canada) | WSP-M2C197-A | 1.15L (2.43 pt)[b] |

[a] Power Transfer Unit (PTU)

[b] Rear Drive Unit (RDU)

**TORQUE SPECIFICATIONS**

| Description | N.m | lb-in |
|---|---|---|
| 4X4 control module nut | 7 | 62 |

## DESCRIPTION AND OPERATION

### FOUR WHEEL DRIVE (4WD) SYSTEMS

The All-Wheel Drive (AWD) system consists of the following:

- Power Transfer Unit (PTU)
- Rear driveshaft
- 4X4 control module (coupling device control module) (located behind the RH side instrument panel)
- Rear axle with coupling device

Torque from the engine is transferred through the transaxle to the PTU. This torque is transferred from the driveshaft to the rear axle, which drives the rear halfshafts. The AWD system, referred to as an Active Torque Coupling (ATC) system, is always active and requires no driver input.

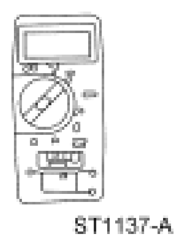
The AWD system continuously monitors vehicle conditions and automatically adjusts the torque distribution between the front and rear wheels. During normal operation, most of the torque is delivered to the front wheels. If wheel slip between the front and rear wheels is detected, or if the vehicle is under heavy acceleration, the AWD system increases torque to the rear wheels to prevent or control wheel slip.

Serviceable components of the PTU are limited to the output shaft seal and flange, intermediate shaft seal and deflector, and the PTU-transaxle compression seal. No internal components are serviced. There should be no need to remove the PTU cover. If any of the internal geared components, bearings, case cover or shafts are worn or damaged, a new PTU must be installed.

# DIAGNOSTIC TESTS

## FOUR WHEEL DRIVE (4WD) SYSTEMS - ALL WHEEL DRIVE (AWD)

### SPECIAL TOOLS

| Illustration | Tool Name | Tool Number |
|---|---|---|
|   | 73III Automotive Meter | 105-R0057 or equivalent |
|   | Flex Probe Kit | 105-R025C |
|   | Vehicle Communication Module (VCM) and Integrated Diagnostic System (IDS) | software with appropriate hardware, or equivalent scan tool |

**Principles of Operation**

The vehicle is equipped with a four wheel drive (4WD) system, **also referred to as active torque control coupling (ATCC) or intelligent torque control coupling (ITCC)** , that is always active and requires no driver input. The system has no mode select switch. The system combines transparent all-surface operation with all wheel drive (AWD), and is capable of handling all road conditions, including street and highway driving as well as off-road and winter driving.

The system continuously monitors vehicle conditions and automatically adjusts the torque distribution between the front and rear wheels. During normal operation, most of the torque is sent to the front wheels. If wheel slip between the front and rear wheels is detected, or if the vehicle is under heavy acceleration (high-throttle position), the AWD system increases torque to the rear wheels to prevent or control wheel slip.

The system consists of a power transfer unit (PTU), 4X4 control module, rear axle and a solenoid-actuated ATCC device. The module varies the torque sent to the rear wheels by controlling the current sent to the clutch located inside the rear axle.

The 4X4 control module also provides the brake system with its current clutch duty cycle and whether or not the brake system may take command of the clutch duty cycle.

**NOTE:** **The ATCC solenoid is not repairable. If a new component is required, the ATCC solenoid and rear axle are installed as an assembly. Refer to REAR DRIVE AXLE/DIFFERENTIAL article.**

The PTU is a gearbox that attaches to the transaxle. The RH intermediate shaft passes through the PTU and engages the differential side gear as in normal front wheel drive (FWD) applications. The transaxle differential drives the PTU. The PTU then drives the driveshaft at all times. The driveshaft drives one half of the rear axle clutch pack. The other half of the rear axle clutch pack drives the rear axle ring and pinion.

The active, on-demand AWD system uses data from other systems as inputs to the 4X4 control module. The 4X4 control module uses the inputs to determine the appropriate amount of current to send to the ATCC solenoid that delivers the desired torque to the rear wheels. Specific inputs to the 4X4 control module are:

- Accelerator pedal position via the high-speed controller area network (HS-CAN)
- Transaxle range from the PCM via the HS-CAN
- Brake system status from the ABS module via the HS-CAN
- Wheel speed from all 4 wheels from the ABS module via the HS-CAN

4X4 control module outputs are:

- Solid-state clutch (pulse-width modulated signal) to the ATCC solenoid
- Percent of torque transfer commanded signal to the ABS module via the HS-CAN
- Torque request available signal to the ABS module via the HS-CAN

**Heat Protection Mode -** During very extreme off-road operation, the AWD system utilizes a heat protection mode to protect the active torque control coupling (ATCC) solenoid (part of rear axle) from damage. If the system detects an overheat condition, it enters a locked mode. If the heat in the system continues to rise once in the locked mode, the 4X4 control module disables the ATCC solenoid. Allow the system to cool down at least 10 minutes with the ignition switch in the ON position.

**Inspection and Verification**

1. Verify the customer concern.
2. Inspect for obvious signs of mechanical or electrical damage.

**VISUAL INSPECTION CHART**

| Mechanical | Electrical |
| --- | --- |
| | |

| | |
|---|---|
| • Active torque control coupling (ATCC) solenoid (part of rear axle)<br>• Power transfer unit (PTU)<br>• Rear drive unit (RDU)<br>• Halfshafts and CV joints<br>• Driveshaft and U-joints<br>• Fluid leaks<br>• Wheel/tire size and brand<br>• Matching tire size and brand<br>• Tire pressure | • Smart junction box (SJB) fuse(s):<br>○ 11 (10A)<br>○ 35 (10A)<br>• 4X4 control module<br>• Wiring harness<br>• Connector(s)<br>• Circuitry |

3. If an obvious cause for an observed or reported concern is found, correct the cause (if possible) before proceeding to the next step.

**NOTE: Make sure to use the latest scan tool software release.**

4. If the cause is not visually evident, connect the scan tool to the data link connector (DLC).

**NOTE: The vehicle communication module (VCM) LED prove out confirms power and ground from the DLC are provided to the VCM.**

5. If the scan tool does not communicate with the VCM:
   - check the VCM connection to the vehicle.
   - check the scan tool connection to the VCM.
   - refer to **MODULE COMMUNICATIONS NETWORK** article, No Power To The Scan Tool, to diagnose no communication with the scan tool.
6. If the scan tool does not communicate with the vehicle:
   - verify the ignition key is in the ON position.
   - refer to **MODULE COMMUNICATIONS NETWORK** article to diagnose no response from the PCM.
7. Carry out the network test.
   - If the scan tool responds with no communication for one or more modules, refer to **MODULE COMMUNICATIONS NETWORK** article.
   - If the network test passes, retrieve and record continuous memory DTCs.
8. Clear the DTCs and carry out the self-test diagnostics for the 4X4 control module.

9. If the DTCs retrieved are related to the concern, go to the 4X4 Control Module DTC Chart. For all other DTCs, refer to **MULTIFUNCTION ELECTRONIC MODULES** article.
10. If no DTCs related to the concern are retrieved, go to **SYMPTOM CHART - FOUR WHEEL DRIVE (4WD) SYSTEMS** table.

**DTC Charts**

**4X4 CONTROL MODULE DTC CHART**

| DTC | Description | Possible Causes | Source | Action |
|---|---|---|---|---|
| B1317 | Battery Voltage High | • Generator | 4X4 Control Module | Go to **Pinpoint Test B** . |
| B1318 | Battery Voltage Low | • Blown fuse (B+)<br>• Intermittent open on power circuit<br>• 4X4 control module connector | 4X4 Control Module | CHECK smart junction box (SJB) fuses 11 (10A) and 35 (10A). If fuses are OK, go to **Pinpoint Test B** . |
| B1342 | ECU is Faulted | • Microprocessor or internal memory fault | 4X4 Control Module | CLEAR the DTCs. REPEAT the self-test. If DTC B1342 is retrieved again, INSTALL a new 4X4 control module. REFER to **4X4 CONTROL MODULE**. |
| B2900 | VIN Mismatch | • Incorrect VIN codes received from PCM | 4X4 Control Module | CHECK the PCM configuration. REFER to **MODULE CONFIGURATION** article. |
| P1635 | Tire/Axle Out of Acceptable Range | • Incorrect size tire installed on vehicle<br>• Wheel speed sensor failure<br>• Flat or under-inflated tire | 4X4 Control Module | Go to **Pinpoint Test G** . |
| P1824 | 4-Wheel Drive Clutch Relay Circuit Failure | • Short to ground on active torque control coupling (ATCC) solenoid (part of rear axle) high circuit<br>• 4X4 control module failure | 4X4 Control Module | CHECK SJB fuse 11 (10A). If the fuse is OK, go to **Pinpoint Test E** . |
| P1825 | 4-Wheel Drive | • 4X4 control | 4X4 Control | CHECK SJB fuses 11 (10A) |

| | | | | |
|---|---|---|---|---|
| | Clutch Relay Open Circuit | module harness connector<br>• Open in ATCC solenoid (part of rear axle) low circuit<br>• Open in ATCC solenoid high circuit<br>• Short to ground on ATCC solenoid low circuit<br>• Short to ground on ATCC solenoid high circuit<br>• Open coil inside ATCC solenoid<br>• Short to voltage in ATCC solenoid low circuit<br>• 4X4 control module failure | Module | and 35 (10A). If the fuses are OK, go to **Pinpoint Test E** . |
| U0100 | Lost Communication with ECM/PCM | • Controller area network (CAN) bus fault<br>• No data received by 4X4 control module from PCM | 4X4 Control Module | Go to **Pinpoint Test D** . |
| U0121 | Lost Communication with Anti-lock Brake System (ABS) Control Module | • CAN bus fault<br>• No data received by 4X4 control module from ABS control module | 4X4 Control Module | Go to **Pinpoint Test D** . |
| U0155 | Lost Communication with Instrument Cluster (IC) Control Module | • High-speed controller area network (HS-CAN) fault<br>• No data received from the IC by the 4X4 control module | 4X4 Control Module | REFER to **MODULE COMMUNICATIONS NETWORK** article to diagnose the lost communication with the IC. |
| U0401 | Invalid Data Received From | • Invalid data | 4X4 Control Module | REFER to **INTRODUCTION -** |

| | | | | |
|---|---|---|---|---|
| | ECM/PCM A | received from PCM<br>• Data out of range | | **GASOLINE ENGINES** article to diagnose the accelerator position, engine coolant or keep alive memory fault. |
| U0415 | Invalid Data Received From Anti-lock Brake System (ABS) Control Module | • Invalid data received from ABS module<br>• Data out of range | 4X4 Control Module | Go to **Pinpoint Test C** . |
| U2050 | No Application Present | • Application software is not present<br>• Application software update failed<br>• 4X4 control module failed | 4X4 Control Module | CONFIGURE the 4X4 control module. REFER to **MODULE CONFIGURATION** article to CARRY OUT programmable module installation (PMI). CLEAR the DTCs. RETRIEVE the DTCs and VERIFY successful module configuration. If DTC U2050 is retrieved again, INSTALL a new 4X4 control module. REFER to **4X4 CONTROL MODULE**. REPEAT the self-test. |
| U2051 | One or More Calibration Files Missing/Corrupt | • Calibration software not present<br>• Calibration software update failed<br>• 4X4 control module failed | 4X4 Control Module | CONFIGURE the 4X4 control module. REFER to **MODULE CONFIGURATION** article to CARRY OUT PMI. CLEAR the DTCs. RETRIEVE the DTCs and VERIFY successful module configuration. If DTC U2050 is retrieved again, INSTALL a new 4X4 control module. REFER to **4X4 CONTROL MODULE**. REPEAT the self-test. |

**Symptom Chart - Four Wheel Drive (4WD) System**

**SYMPTOM CHART - FOUR WHEEL DRIVE (4WD) SYSTEM**

| Condition | Possible Sources | Action |
|---|---|---|

| | | |
|---|---|---|
| • The concern is unverifiable; no DTCs present | • The concern description is inaccurate | • CARRY OUT the All Wheel Drive (AWD) System Functional Test. Go to **Pinpoint Test A** . |
| • The power transfer unit (PTU) makes noise | • Tire inflation pressure<br>• Tire and wheel size | • MAKE SURE all tires and wheels are the same size and brand and the inflation pressures are correct. |
| | • Fluid level | • FILL with the correct type and amount of lubricant. REFER to **TRANSFER CASE - POWER TRANSFER UNIT (PTU)** article. |
| | • Internal components | • OPERATE the vehicle in all gears. If there is noise in the transaxle in NEUTRAL, or in some gears and not in others, REMOVE and REPAIR the transaxle. REFER to **AUTOMATIC TRANSAXLE/TRANSMISSION - 6F50** article. If there is noise in all gears, INSTALL a new PTU. REFER to **TRANSFER CASE - POWER TRANSFER UNIT (PTU)** article. |
| • 4X4 control module battery voltage is high or low | • Concern with voltage regulator or generator<br>• Smart junction box (SJB) fuse 11 (10A)<br>• SJB fuse 35 (10A)<br>• Intermittent open power circuit<br>• 4X4 control module internal relay failure | • Go to **Pinpoint Test B** . |
| • 4X4 control module received no/invalid data from PCM | • Invalid data received from PCM<br>• Data out of range | • REFER to **INTRODUCTION - GASOLINE ENGINES** article to diagnose the accelerator position, engine coolant or keep alive memory fault. |
| • 4X4 control module received no/invalid data from ABS module | • Invalid data received from the ABS module<br>• Data out of range | • Go to **Pinpoint Test C** . |
| • High-speed controller | • HS-CAN bus fault | • Go to **Pinpoint Test D** . |

| | | |
|---|---|---|
| area network (HS-CAN) communications bus fault | • PCM fault<br>• ABS fault<br>• Instrument cluster (IC) fault<br>• 4X4 control module | |
| • Vehicle has no or inadequate torque at rear wheels | • Rear axle | • REFER to **REAR DRIVE AXLE/DIFFERENTIAL** article. |
| | • PTU mechanical failure | • REFER to **TRANSFER CASE - POWER TRANSFER UNIT (PTU)** article. |
| | • Wheels/tires<br>• Active torque control coupling (ATCC) solenoid (part of rear axle)<br>• SJB fuse(s):<br>  • 11 (10A)<br>  • 35 (10A)<br>• 4X4 control module | • Go to **Pinpoint Test G** . |
| | • Circuitry | • Go to **Pinpoint Test E** . |
| • Vehicle binds in a turn or resists turning/pulsates or shudders in a straight line | • Wheels/tires | • REFER to **WHEELS AND TIRES** article. |
| | • Brake system | • REFER to **BRAKE SYSTEM - GENERAL INFORMATION** article. |
| | • Wheel bearings | • REFER to **SUSPENSION SYSTEM - GENERAL INFORMATION** article. |
| | • Halfshafts | • REFER to **DRIVELINE SYSTEM - GENERAL INFORMATION** article. |
| | • Wheel speed sensor(s) | • REFER to **VEHICLE DYNAMIC SYSTEMS** article. |
| | • ABS module | • REFER to **VEHICLE DYNAMIC SYSTEMS** article. |
| | • 4X4 control module<br>• ATCC solenoid | • Go to **Pinpoint Test F** . |

| | | |
|---|---|---|
| | (part of rear axle)<br>• Circuitry | |
| • Tire/axle out of acceptable operating range | • Wheels/tires<br>• Wheel speed sensors<br>• ABS module<br>• 4X4 control module | • Go to **Pinpoint Test G** . |

**Pinpoint Tests**

### Pinpoint Test A: All Wheel Drive (AWD) System Functional Test

Refer to appropriate SYSTEM WIRING DIAGRAMS article for All Wheel Drive (AWD) schematic and connector information.

**Normal Operation**

The all wheel drive (AWD) system continuously monitors vehicle conditions and automatically adjusts the torque distribution between the front and rear wheels. During normal operation, most of the torque is delivered to the front wheels. If wheel slip between the front and rear wheels is detected, or if the vehicle is under heavy acceleration, the AWD system increases torque to the rear wheels to prevent or control wheel slip.

**This pinpoint test is intended to diagnose the following:**

- An AWD system concern without on-demand or continuous DTCs present.

**PINPOINT TEST A: AWD SYSTEM FUNCTIONAL TEST**

**NOTE:** **This test must be carried out on a hard surface in an area without traffic.**

**NOTE:** **Carry out Inspection and Verification as outlined before carrying out this pinpoint test.**

**A1 CHECK FOR WRENCH LIGHT PROVE-OUT**

- Turn the ignition to the ON position and observe the wrench light.
- **Does the wrench light illuminate for 3 seconds and then turn OFF?**

  **YES** : Go to A2.

  **NO** : REFER to **INSTRUMENT CLUSTER (IC), MESSAGE CENTER, AND WARNING CHIMES** article for further diagnosis of the instrument cluster (IC).

**A2 CHECK FOR ACTIVE TORQUE CONTROL COUPLING (ATCC) SOLENOID (PART OF REAR AXLE) LOCK**

- Drive the vehicle on a dry, hard surface in turns while applying the accelerator pedal.

- **Is driveline wind-up present in turns?**

  **YES** : With the ignition switch in ON position, allow the ATCC solenoid to cool down for at least 10 minutes and repeat the test. CHECK for wind-up again. If no wind-up is found, go to A3; if still present, go to **Pinpoint Test F** .

  **NO** : Go to A3.

**A3 CHECK FOR VALID ACCELERATOR PEDAL POSITION MONITORING THE ACCELERATOR PEDAL POSITION (AP) PID**

- Enter the following diagnostic mode on the diagnostic tool: DataLogger - 4X4 Control Module
- Monitor the AP PID while pressing the accelerator pedal.
- **Does the accelerator pedal position match the AP PID percent value?**

  **YES** : Go to A4.

  **NO** : REFER to the INTRODUCTION - GASOLINE ENGINES article to diagnose the accelerator pedal position sensor concern.

**A4 CHECK 4X4 CONTROL MODULE WHEEL SPEED SENSOR PIDs**

- Enter the following diagnostic mode on the diagnostic tool: DataLogger - 4X4 Control Module
- While driving the vehicle at 48 km/h (30 mph), monitor the following wheel speed sensor PIDs:
  - Left Front Wheel Speed Sensor (LF_WSPD)
  - Left Rear Wheel Speed Sensor (LR_WSPD)
  - Right Front Wheel Speed Sensor (RF_WSPD)
  - Right Rear Wheel Speed Sensor (RR_WSPD)
- **Are all 4 wheel speeds within 3 km/h (1.8 mph) of each other?**

  **YES** : Go to A5.

  **NO** : Go to **Pinpoint Test G** .

**A5 CHECK VEHICLE ACCELERATION IN A STRAIGHT LINE**

- Carry out 3 accelerations from 0-48 km/h (0-30 mph) in a straight line (one each with low, medium and full accelerator pedal application).
- **Does the vehicle pulsate or shudder while accelerating?**

  **YES** : Go to **Pinpoint Test F** .

  **NO** : Go to A6.

**A6 CHECK VEHICLE TURNING ABILITY**

- Drive the vehicle in a fully locked turn, on dry pavement, at 8 km/h (5 mph).
- **Does the vehicle bind in the turn or resist turning?**

  **YES** : Go to **Pinpoint Test F** .

  **NO** : Go to A7.

**A7 CHECK FOR TORQUE AT REAR WHEELS USING PWM OUTPUT CONTROL COMMAND #1 (CLCH_SOL) ACTIVE COMMAND**

- Enter the following diagnostic mode on the diagnostic tool: DataLogger - 4X4 Control Module
- Using the scan tool, command CLCH_SOL to energize the ATCC solenoid to a constant 100% applied.

- Drive the vehicle in a fully locked turn, on dry pavement, at 8 km/h (5 mph).
- **Does the vehicle bind in the turn or resist turning?**

  **YES** : End the active command. The concern cannot be duplicated at this time. RETURN the vehicle to the customer.

  **NO** : End the active command. CHECK that the driveshaft rotates when the transaxle is in gear. If yes, go to **Pinpoint Test E** . If no, CHECK the PTU. REFER to **TRANSFER CASE - POWER TRANSFER UNIT (PTU)** article.

**Pinpoint Test B: 4X4 Control Module Battery Voltage is High or Low**

Refer to appropriate SYSTEM WIRING DIAGRAMS article for All Wheel Drive (AWD) schematic and connector information.

**Normal Operation**

If the 4X4 control module observes an overpower or underpower voltage condition, DTCs are set and the 4X4 control module may not allow 4WD operation.

- DTC B1317 Battery Voltage High - When the 4X4 control module detects battery voltage greater than 16 volts for at least 10 seconds or ignition run/start voltage greater than 16 volts for at least one second, this DTC is set.
- DTC B1318 Battery Voltage Low - When the 4X4 control module detects battery voltage less than 9 volts for at least 10 seconds or ignition run/start voltage less than 9 volts for at least one second with the engine running at 1,500 RPM or greater, this DTC is set.

**This pinpoint test is intended to diagnose the following:**

- Fuses
- Charging/battery system
- 4X4 control module
- Wiring, terminals or connectors

**PINPOINT TEST B: 4X4 CONTROL MODULE BATTERY VOLTAGE IS HIGH OR LOW**

**NOTE:** **Carry out Inspection and Verification as outlined before carrying out this pinpoint test.**

**B1 CHECK FOR HIGH BATTERY VOLTAGE MONITORING THE MODULE SUPPLY VOLTAGE (VBAT_4X4) PID**

- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: DataLogger - 4X4 Control Module
- Monitor the battery voltage VBAT_4X4 PID.
- **Is the battery voltage greater than 16 volts?**

**YES** : REFER to **CHARGING SYSTEM - GENERAL INFORMATION** article for diagnosis of the battery and charging system.

**NO** : Go to B2.

**B2 CHECK FOR LOW VOLTAGE MONITORING THE MODULE SUPPLY VOLTAGE (VBAT_4X4) PID**

- Enter the following diagnostic mode on the diagnostic tool: DataLogger - 4X4 Control Module
- Monitor the battery voltage VBAT_4X4 PID.
- **Is the battery voltage less than 9 volts?**

  **YES** : Go to B4.

  **NO** : Go to B3.

**B3 CHECK FOR A BLOWN FUSE**

- Check the following smart junction box (SJB) fuses:
  - 11 (10A).
  - 35 (10A).
- **Are the fuses open?**

  **YES** : INSTALL a new fuse(s). CLEAR the DTCs. REPEAT the self-test.

  **NO** : Go to B4.

**B4 CHECK FOR CORRECT VOLTAGE AT THE BATTERY**

- Measure the voltage between the battery positive terminal and ground.
- **Is the voltage greater than 10 volts?**

  **YES** : Go to B5.

  **NO** : REFER to **CHARGING SYSTEM - GENERAL INFORMATION** article for diagnosis of the battery and charging system.

**B5 CHECK CIRCUIT GD140 (BK/GN) FOR AN OPEN**

- Key in OFF position.
- Disconnect: 4X4 Control Module C281
- Measure the resistance between 4X4 control module C281-15, circuit GD140 (BK/GN), harness side and ground.

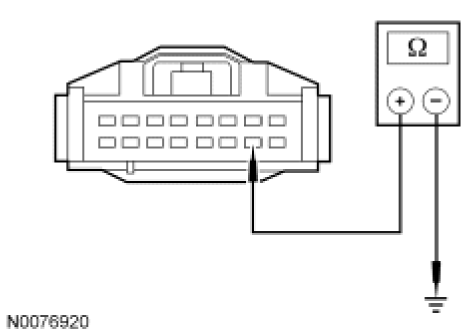



**Fig. 1: Checking Circuit GD126 (BK/WH) For An Open**
**Courtesy of FORD MOTOR CO.**

- Repeat this measurement while wiggling the harness.
- **Is the resistance less than 5 ohms?**

  **YES** : Go to B6.

  **NO** : REPAIR the circuit. CLEAR the DTCs. REPEAT the self-test.

**B6 CHECK POWER TO THE 4X4 CONTROL MODULE**

- Key in ON position.

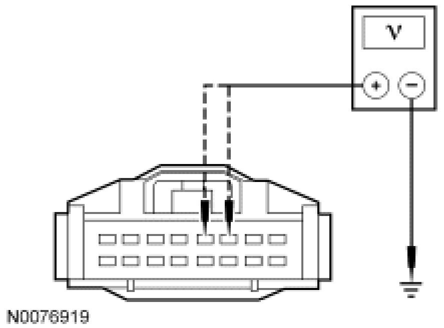



**Fig. 2: Checking 4X4 Control Module Power Circuits**
**Courtesy of FORD MOTOR CO.**

- Measure the voltage between 4X4 control module C281-6, circuit SBP11 (BU/RD), harness side and ground; and between 4X4 control module C281-5, circuit CBP35 (YE/GY), harness side and ground.
- **Are the voltages greater than 10 volts?**

  **YES** : Go to B7.

  **NO** : REPAIR the circuit in question. CLEAR the DTCs. REPEAT the self-test.

**B7 CHECK FOR CORRECT 4X4 CONTROL MODULE OPERATION**

- Disconnect the 4X4 control module.
- Check the harness and component side connectors for:
  - corrosion.
  - pushed-out/bent pins.
- Connect the 4X4 control module and make sure it seats correctly.
- Operate the system and determine if the concern is still present.
- **Is the concern still present?**

  **YES** : INSTALL a new 4X4 control module. REFER to **4X4 CONTROL MODULE**. CLEAR the DTCs. REPEAT the self-test.

  **NO** : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. CLEAR the DTCs. REPEAT the self-test.

**Pinpoint Test C: 4X4 Control Module Received No/Invalid Data From ABS Module**

Refer to appropriate SYSTEM WIRING DIAGRAMS article for All Wheel Drive (AWD) schematic and connector information.

**Normal Operation**

The on-demand four wheel drive (4WD) system uses data from other systems as inputs to the 4X4 control module. The 4X4 control module uses the inputs to determine the appropriate duty cycle to send to the active torque control coupling (ATCC) (part of rear axle) solenoid that delivers the desired torque to the rear wheels. Specific inputs to the 4X4 control module are: accelerator pedal position output from the PCM, transmission range from the PCM, brake system status and all 4 wheel speeds from the ABS module. Communication between the 4X4 control module and other modules is obtained through the high-speed controller area network (HS-CAN). If the 4X4 control module loses communication with, or receives invalid data from any of the necessary modules, DTCs are set and the 4X4 control module may not allow 4WD operation.

- DTC U0415 Invalid Data Received from Anti-lock Brake System (ABS) Control Module - When the 4X4 control module receives invalid HS-CAN bus data or invalid wheel speed(s) from the ABS module for more than 30 seconds, this DTC is set.

**This pinpoint test is intended to diagnose the following:**

- ABS module

**PINPOINT TEST C: 4X4 CONTROL MODULE RECEIVED NO/INVALID DATA FROM ABS MODULE**

| | |
|---|---|
| **NOTE:** | **This test must be carried out on a hard surface in an area without traffic.** |

| | |
|---|---|
| **NOTE:** | **Carry out Inspection and Verification as outlined before carrying out this pinpoint test.** |

**C1 CHECK FOR ABS MODULE DTCs**

- Key in ON position.
- Carry out the ABS module self-test.
- Review the results of the ABS module self-test.
- **Are any DTCs present?**

  **YES** : REFER to **VEHICLE DYNAMIC SYSTEMS** article for further diagnosis of the ABS.

  **NO** : Go to C2.

**C2 CHECK WHEEL SPEED SENSORS PIDs**

- Enter the following diagnostic mode on the diagnostic tool: DataLogger - 4X4 Control Module
- While driving the vehicle at 32 km/h (20 mph), monitor the following wheel speed sensor PIDs:
  - Left Front Wheel Speed Sensor (LF_WSPD)
  - Left Rear Wheel Speed Sensor (LR_WSPD)
  - Right Front Wheel Speed Sensor (RF_WSPD)
  - Right Rear Wheel Speed Sensor (RR_WSPD)

- **Do the 4 wheel speed sensors match the speedometer and are they within 3 km/h (1.8 mph) of each other?**

  **YES** : The system is operating correctly at this time. CLEAR the DTCs. REPEAT the self-test.

  **NO** : REFER to **VEHICLE DYNAMIC SYSTEMS** article for further diagnosis of the ABS.

**Pinpoint Test D: High-Speed Controller Area Network (HS-CAN) Communications Bus Fault**

Refer to appropriate SYSTEM WIRING DIAGRAMS article for All Wheel Drive (AWD) schematic and connector information.

**Normal Operation**

The on-demand four wheel drive (4WD) system uses data from other systems as inputs to the 4X4 control module. The 4X4 control module uses the inputs to determine the appropriate duty cycle to send to the active torque control coupling (ATCC) solenoid (part of rear axle) that delivers the desired torque to the rear wheels. Specific inputs to the 4X4 control module are: accelerator pedal position output from the PCM, transmission range from the PCM, brake system status and all 4 wheel speeds from the ABS module. Communication between the 4X4 control module and other modules is obtained through the high-speed controller area network (HS-CAN). If the 4X4 control module loses communication with, or receives invalid data from any of the necessary modules, DTCs are set and the 4X4 control module may not allow 4WD operation.

- DTC U0100 Lost Communication with ECM/PCM - When the 4X4 control module detects a missing HS-CAN bus message from the PCM for at least 5 seconds, this DTC is set.
- DTC U0121 Lost Communication with Anti-lock Brake System (ABS) Control Module - When the 4X4 control module detects a missing HS-CAN bus message from the ABS module for at least 5 seconds, this DTC is set.

**This pinpoint test is intended to diagnose the following:**

- 4X4 control module
- PCM
- ABS module
- Wiring, terminals or connectors

**PINPOINT TEST D: HIGH-SPEED CONTROLLER AREA NETWORK (HS-CAN) COMMUNICATIONS BUS FAULT**

**NOTE:** **Carry out Inspection and Verification as outlined before carrying out this pinpoint test.**

**D1 CHECK HS-CAN CIRCUITS BETWEEN THE DATA LINK CONNECTOR (DLC) AND THE 4X4 CONTROL MODULE FOR AN OPEN**

- Key in OFF position.

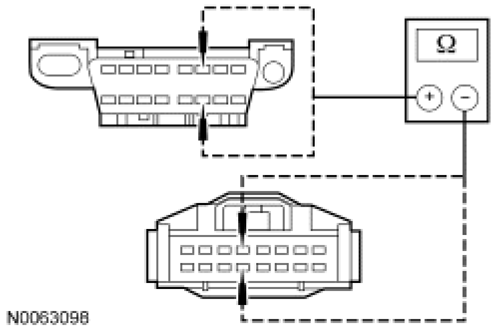



**Fig. 3: Checking High Speed Can Circuits Between DLC And 4X4 Control Module For An Open**
**Courtesy of FORD MOTOR CO.**

- Measure the resistance between 4X4 control module C281-4, circuit VDB04 (WH/BU), harness side and DLC C251-6, circuit VDB04 (WH/BU), harness side; and between 4X4 control module C281-12, circuit VDB05 (WH), harness side and DLC C251-14, circuit VDB05 (WH), harness side.
- **Are the resistances less than 5 ohms?**

  **YES** : Go to D2.

  **NO** : REPAIR the circuit(s). CLEAR the DTCs. REPEAT the self-test. CARRY OUT the Network Test.

**D2 CHECK HS-CAN CIRCUITS FOR A SHORT TO GROUND**

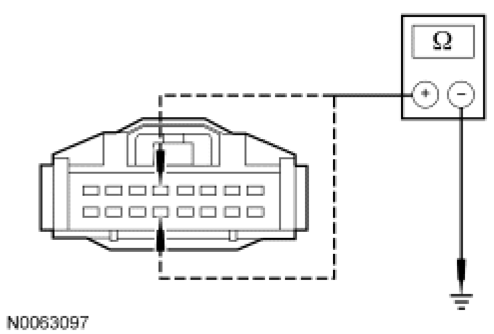



**Fig. 4: Checking High Speed CAN Circuits For A Short To Ground**
**Courtesy of FORD MOTOR CO.**

- Measure the resistance between 4X4 control module C281-4, circuit VDB04 (WH/BU), harness side and ground; and between 4X4 control module C281-12, circuit VDB05 (WH), harness side and ground.
- **Are the resistances greater than 5,000 ohms?**

  **YES** : Go to D3.

  **NO** : REPAIR the circuit(s). CLEAR the DTCs. REPEAT the self-test. CARRY OUT the Network Test.

**D3 CHECK RESISTANCE BETWEEN HS-CAN CIRCUITS**

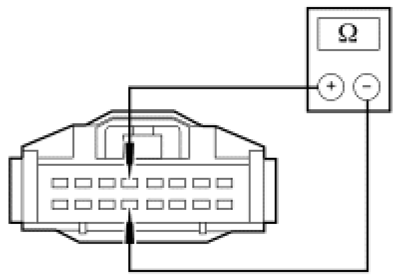



**Fig. 5: Checking Resistance Between High Speed CAN Circuits**
**Courtesy of FORD MOTOR CO.**

| | |
|---|---|
| **NOTE:** | **The PCM and the instrument cluster (IC) must be connected and operating correctly to obtain the correct resistance between the HS-CAN circuits.** |

- Measure the resistance between 4X4 control module C281-4, circuit VDB04 (WH/BU) harness side and 4X4 control module C281-12, circuit VDB05 (WH) harness side.
- **Is the resistance 60 ohms (± 5 ohms)?**

**YES** : Go to D4.

**NO** : REPAIR the circuit(s). CLEAR the DTCs. REPEAT the self-test. CARRY OUT the Network Test.

**D4 CHECK FOR DTCs U0100 AND U0121 SET TOGETHER**

- Key in ON position.
- Carry out the 4X4 control module self-test.
- Review the results of the 4X4 control module self-test.
- **Are DTCs U0100 and U0121 set together?**

**YES** : REFER to **MODULE COMMUNICATIONS NETWORK** article to continue diagnosis of the communications network.

**NO** : Go to D5.

**D5 CHECK FOR DTC U0100**

- Refer to the results of the 4X4 control module self-test.
- **Is DTC U0100 set?**

**YES** : The PCM is not sending any data to the 4X4 control module. REFER to **MODULE COMMUNICATIONS NETWORK** article to for additional PCM diagnosis.

**NO** : Go to D6.

**D6 CHECK FOR DTC U0121**

- Refer to the results of the 4X4 control module self-test.
- **Is DTC U0121 set?**

**YES** : The ABS module is not sending any data to the 4X4 control module. REFER to **MODULE COMMUNICATIONS NETWORK** article for additional ABS module diagnosis.

**NO** : Go to D7.

**D7 CHECK FOR CORRECT 4X4 CONTROL MODULE OPERATION**

- Key in OFF position.
- Disconnect the 4X4 control module.
- Check the harness and component side connectors for:
  - corrosion.
  - pushed-out/bent pins.
- Connect the 4X4 control module and make sure it seats correctly.
- Operate the system and determine if the concern is still present.
- **Is the concern still present?**

  **YES** : INSTALL a new 4X4 control module. REFER to **4X4 CONTROL MODULE**. CLEAR the DTCs. REPEAT the self-test.

  **NO** : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. CLEAR the DTCs. REPEAT the self-test.

**Pinpoint Test E: Vehicle Has No or Inadequate Torque at Rear Wheels**

Refer to appropriate SYSTEM WIRING DIAGRAMS article for All Wheel Drive (AWD) schematic and connector information.

**Normal Operation**

The on-demand four wheel drive (4WD) system uses data from other systems as inputs to the 4X4 control module. The 4X4 control module uses the inputs to determine the appropriate duty cycle to send to the active torque control coupling (ATCC) solenoid (part of rear axle) that delivers the desired torque to the rear wheels. If the 4X4 control module loses communication with, or receives invalid data from any of the necessary modules, DTCs are set and the 4X4 control module may not allow 4WD operation.

- DTC P1824 4-Wheel Drive Clutch Relay Circuit Failure - When the 4X4 control module detects a short to ground on the ATCC solenoid feedback circuit, this DTC is set.
- DTC P1825 4-Wheel Drive Clutch Relay Open Circuit - When the 4X4 control module detects an open or short to ground on the ATCC solenoid command or feedback circuit for more than 2 seconds, this DTC is set.

**This pinpoint test is intended to diagnose the following:**

- Tire/axle out of range
- ATCC solenoid (part of rear axle)
- 4X4 control module
- Power transfer unit (PTU)
- Wiring, terminals or connectors

## PINPOINT TEST E: VEHICLE HAS NO OR INADEQUATE TORQUE AT REAR WHEELS

**NOTE:** **Carry out Inspection and Verification as outlined before carrying out this pinpoint test.**

### E1 CHECK FOR 4X4 CONTROL MODULE DTCs

- Key in ON position.
- Carry out the 4X4 control module self-test.
- **Are any DTCs received?**

  **YES** : For DTC P1824 or P1825, go to E5.

  For all other DTCs, REFER to the 4X4 Control Module DTC Chart in this service information.

  **NO** : Go to E2.

### E2 CHECK HIGH-SPEED CONTROLLER AREA NETWORK (HS-CAN) CIRCUITS BETWEEN DATA LINK CONNECTOR (DLC) AND 4X4 CONTROL MODULE FOR AN OPEN

- Key in OFF position.
- Disconnect: 4X4 Control Module C281

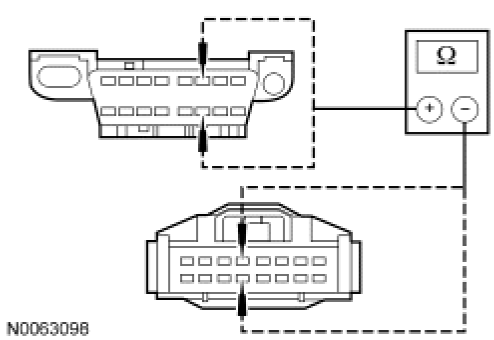



**Fig. 6: Checking High Speed Can Circuits Between DLC And 4X4 Control Module For An Open**
**Courtesy of FORD MOTOR CO.**

- Measure the resistance between 4X4 control module C281-4, circuit VDB04 (WH/BU), harness side and DLC C251-6, circuit VDB04 (WH/BU), harness side; and between 4X4 control module C281-12, circuit VDB05 (WH), harness side and DLC C251-14, circuit VDB05 (WH), harness side.
- **Are the resistances less than 5 ohms?**

  **YES** : Go to E3.

  **NO** : REPAIR the circuit(s). CLEAR the DTCs. REPEAT the self-test. CARRY OUT the Network Test.

### E3 CHECK HS-CAN CIRCUITS FOR A SHORT TO GROUND

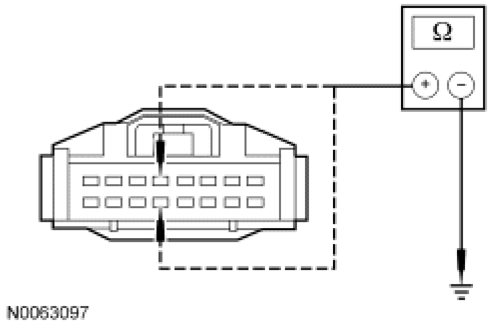



**Fig. 7: Checking High Speed CAN Circuits For A Short To Ground**
**Courtesy of FORD MOTOR CO.**

- Measure the resistance between 4X4 control module C281-4, circuit VDB04 (WH/BU), harness side and ground; and between 4X4 control module C281-12, circuit VDB05 (WH), harness side and ground.
- **Are the resistances greater than 5,000 ohms?**

  **YES** : Go to E4.

  **NO** : REPAIR the circuit(s). CLEAR the DTCs. REPEAT the self-test. CARRY OUT the Network Test.

**E4 CHECK RESISTANCE BETWEEN HS-CAN CIRCUITS**

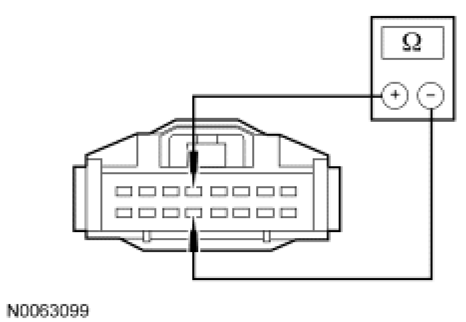



**Fig. 8: Checking Resistance Between High Speed CAN Circuits**
**Courtesy of FORD MOTOR CO.**

| | |
|---|---|
| **NOTE:** | **The PCM and the instrument cluster (IC) must be connected and operating correctly to obtain the correct resistance between the HS-CAN circuits.** |

- Measure the resistance between 4X4 control module C281-4, circuit VDB04 (WH/BU) harness side and 4X4 control module C281-12, circuit VDB05 (WH) harness side.
- **Is the resistance 60 ohms (± 5 ohms)?**

  **YES** : Go to E5.

  **NO** : REPAIR the circuit(s). CLEAR the DTCs. REPEAT the self-test. CARRY OUT the Network Test.

**E5 CHECK CIRCUITS CCF21 (VT/WH) AND RCF21 (WH/VT) FOR A SHORT TO GROUND WITH ATCC SOLENOID (PART OF REAR AXLE) CONNECTED**

- Key in OFF position.
- Disconnect: 4X4 Control Module C281

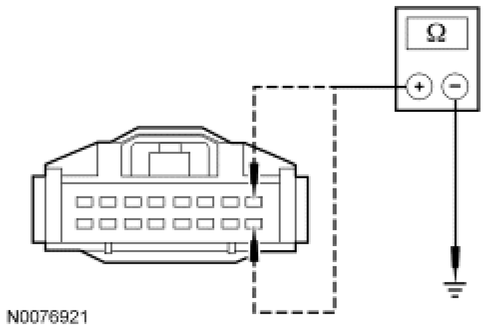



**Fig. 9: Checking Circuits CCF21 (VT/WH) & RCF21 (WH/VT) For A Short To Ground With ATC Solenoid Connected**
**Courtesy of FORD MOTOR CO.**

- Measure the resistance between 4X4 control module C281-8, circuit CCF21 (VT/WH), harness side and ground; and between 4X4 control module C281-16, circuit RCF21 (WH/VT), harness side and ground.
- **Are the resistances greater than 10,000 ohms?**

  **YES** : Go to E8.

  **NO** : Go to E6.

**E6 CHECK CIRCUITS CCF21 (VT/WH) AND RCF21 (WH/VT) FOR A SHORT TO GROUND WITH C4004 DISCONNECTED**

- Disconnect: ATCC Solenoid Jumper Harness C4004

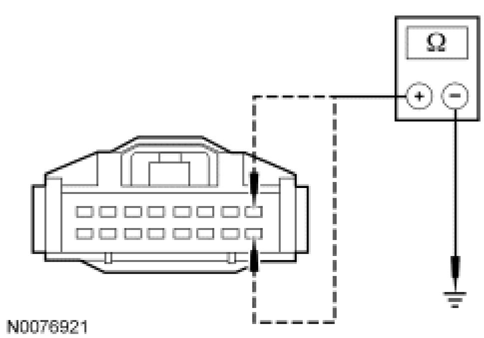



**Fig. 10: Checking Circuits CCF21 (VT/WH) & RCF21 (WH/VT) For A Short To Ground With ATC Solenoid Connected**
**Courtesy of FORD MOTOR CO.**

- Measure the resistance between 4X4 control module C281-8, circuit CCF21 (VT/WH), harness side and ground; and between 4X4 control module C281-16, circuit RCF21 (WH/VT), harness side and ground.

- **Are the resistances greater than 10,000 ohms?**

  **YES** : Go to E7.

  **NO** : REPAIR the circuit(s) in question. CLEAR the DTCs. REPEAT the self-test.

**E7 CHECKING THE ATCC SOLENOID JUMPER HARNESS WIRES**

| | |
|---|---|
| **NOTE:** | **The active torque control coupling (ATCC) solenoid jumper harness must be secured to prevent damage to the wiring.** |

- Disconnect: ATCC Solenoid C4347
- Inspect the ATCC solenoid jumper harness for damage.
- **Is the ATCC solenoid jumper harness damaged?**

  **YES** : REPAIR or INSTALL a new ATCC solenoid jumper harness. CLEAR the DTCs. REPEAT the self-test.

  **NO** : INSTALL a new rear axle assembly. REFER to **REAR DRIVE AXLE/DIFFERENTIAL** article. CLEAR the DTCs. REPEAT the self-test.

**E8 CHECK CIRCUITS CCF21 (VT/WH) AND RCF21 (WH/VT) FOR AN OPEN WITH ATCC SOLENOID CONNECTED**

- Connect: ATCC Solenoid C4347 and ATCC Solenoid Jumper Harness C4004

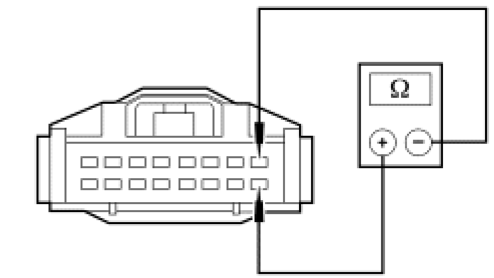



**Fig. 11: Checking Circuits CCF21 (VT/WH) & RCF21 (WH/VT) For An Open With ATC Solenoid Connected**
**Courtesy of FORD MOTOR CO.**

- Measure the resistance between 4X4 control module C281-8, circuit CCF21 (VT/WH), harness side and 4X4 control module C281-16, circuit RCF21 (WH/VT), harness side.
- **Is the resistance less than 10 ohms?**

  **YES** : Go to E11.

  **NO** : Go to E9.

**E9 CHECK CIRCUITS CCF21 (VT/WH) AND RCF21 (WH/VT) FOR AN OPEN AT C4004**

- Disconnect: ATCC Solenoid Jumper Harness C4004

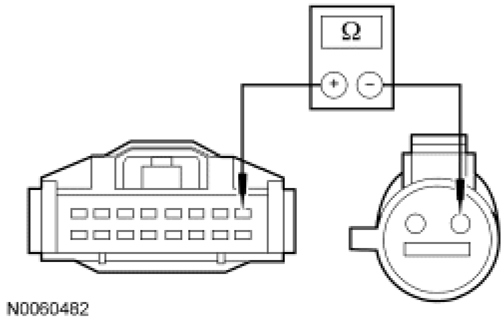



**Fig. 12: Measuring Resistance Between 4X4 Control Module C281-8, Circuit CCF21 (VT/WH) & ATCC solenoid jumper harness C4004-1, circuit CCF21 (VT/WH)**
**Courtesy of FORD MOTOR CO.**

- Measure the resistance between 4X4 control module C281-8, circuit CCF21 (VT/WH), harness side and ATCC solenoid jumper harness C4004-1, circuit CCF21 (VT/WH).
- Repeat this measurement while wiggling the harness.

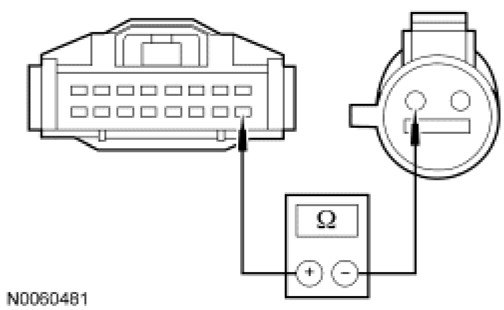



**Fig. 13: Measuring Resistance Between 4X4 Control Module C281-16, Circuit RCF21 (WH/VT) & ATCC Solenoid Jumper Harness C4004-2, Circuit RCF21 (WH/VT)**
**Courtesy of FORD MOTOR CO.**

- Measure the resistance between 4X4 control module C281-16, circuit RCF21 (WH/VT), harness side and ATCC solenoid jumper harness C4004-2, circuit RCF21 (WH/VT), harness side.
- Repeat this measurement while wiggling the harness.
- **Are the resistances less than 5 ohms?**

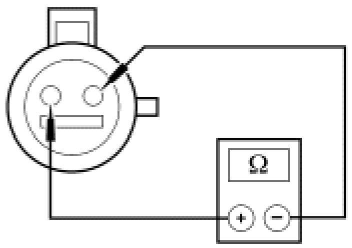
  **YES** : Go to E10.

  **NO** : REPAIR the circuit in question. CLEAR the DTCs. REPEAT the self-test.

**E10 CHECK THE ATCC SOLENOID CIRCUIT FOR AN INTERNAL OPEN**

**NOTE:** **The active torque control coupling (ATCC) solenoid jumper harness must be secured to prevent damage to the wiring.**

- Disconnect: ATCC Solenoid C4347




N0076923

**Fig. 14: Checking ATC Solenoid Circuit For An Internal Open**
**Courtesy of FORD MOTOR CO.**

- Measure the resistance between ATCC solenoid C4347-1, circuit CCF21 (VT/WH), component side and ATCC solenoid C4347-2, circuit RCF21 (WH/VT), component side.
- Repeat this measurement while wiggling the harness.
- **Is the resistance less than 10 ohms?**

  **YES** : REPAIR or INSTALL a new ATCC solenoid jumper harness. CLEAR the DTCs. REPEAT the self-test.

  **NO** : INSTALL a new rear axle assembly. REFER to **REAR DRIVE AXLE/DIFFERENTIAL** article. CLEAR the DTCs. REPEAT the self-test.

**E11 CHECK CIRCUIT GD140 (BK/GN) FOR AN OPEN**

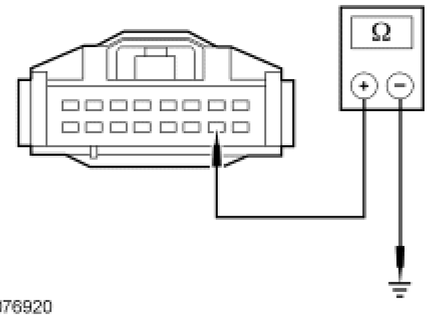



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**Fig. 15: Checking Circuit GD126 (BK/WH) For An Open**
**Courtesy of FORD MOTOR CO.**

- Measure the resistance between 4X4 control module C281-15, circuit GD140 (BK/GN), harness side and ground.
- Repeat this measurement while wiggling the harness.
- **Is the resistance less than 5 ohms?**

  **YES** : Go to E12.

  **NO** : REPAIR the circuit. CLEAR the DTCs. REPEAT the self-test.

**E12 CHECK THE 4X4 CONTROL MODULE POWER CIRCUITS**

- Key in ON position.

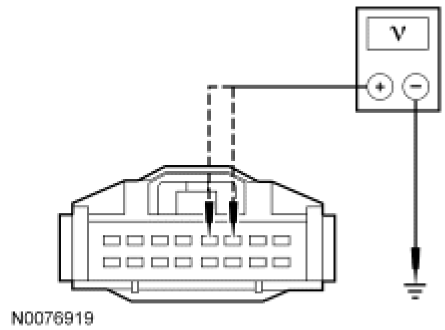


**Fig. 16: Checking 4X4 Control Module Power Circuits**
**Courtesy of FORD MOTOR CO.**

- Measure the voltage between 4X4 control module C281-6, circuit SBP11 (BU/RD), harness side and ground; and between 4X4 control module C281-5, circuit CBP35 (YE/GY), harness side and ground.
- **Are the voltages greater than 10 volts?**

  **YES** : Go to E13.

  **NO** : REPAIR the circuit in question. CLEAR the DTCs. REPEAT the self-test.

**E13 CHECK FOR CORRECT 4X4 CONTROL MODULE OPERATION**

- Disconnect: 4X4 Control Module C281
- Check the harness and component side connectors for:
  - corrosion.
  - pushed-out/bent pins.
- Check the ATCC solenoid jumper harness between the 4X4 control module and the ATCC solenoid, located on the rear subframe.
- Connect the 4X4 control module and make sure it seats correctly.
- Operate the system and determine if the concern is still present.
- **Is the concern still present?**

  **YES** : INSTALL a new 4X4 control module. REFER to **4X4 CONTROL MODULE**. CLEAR the DTCs. REPEAT the self-test.

  **NO** : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. CLEAR the DTCs. REPEAT the self-test.

### Pinpoint Test F: Vehicle Binds in a Turn or Resists Turning/Pulsates or Shudders in a Straight Line

Refer to appropriate SYSTEM WIRING DIAGRAMS article for All Wheel Drive (AWD) schematic and connector information.

### Normal Operation

The all wheel drive (AWD) system continuously monitors vehicle conditions and automatically adjusts the torque distribution between the front and rear wheels. During normal operation, most of the torque is sent to the front wheels. If wheel slip between the front and rear wheels is detected or the vehicle is under heavy acceleration, the AWD system increases torque to the rear wheels.

**This pinpoint test is intended to diagnose the following:**

- Rear axle
- Wheels/tires
- PCM
- ABS module
- 4X4 control module
- Circuitry

**PINPOINT TEST F: VEHICLE BINDS IN A TURN OR RESISTS TURNING/PULSATES OR SHUDDERS IN A STRAIGHT LINE**

**NOTE:** **Carry out Inspection and Verification as outlined before carrying out this pinpoint test.**

**F1 CHECK FOR TORQUE AT THE REAR WHEELS**

- Key in OFF position.
- Disconnect: Active Torque Control Coupling (ATCC) Solenoid (Part of Rear Axle) C4347
- Drive the vehicle in a straight line on dry pavement.
- **Is a pulsation or shudder still present?**

  **YES** : REFER to **DRIVELINE SYSTEM - GENERAL INFORMATION** article for additional axle diagnosis.

  **NO** : Go to F2.

**F2 CHECK TO SEE IF THE ATCC SOLENOID IS BEING COMMANDED ON MONITORING 4WD CLUTCH PWM STATUS (4WDCPWMST) PID**

- Key in OFF position.
- Connect: ATCC Solenoid C4347
- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: DataLogger - 4X4 Control Module
- Monitor the 4WDCPWMST PID.
- Drive the vehicle no faster than 8 km/h (5 mph) in tight turns on dry pavement while monitoring the ATCC solenoid duty cycle.
- **Is the duty cycle greater than 20%?**

  **YES** : Go to F3.

  **NO** : Go to **Pinpoint Test G** .

**F3 CHECK FOR THE CORRECT WHEEL SPEEDS USING THE WHEEL SPEED SENSORS PIDs**

- Enter the following diagnostic mode on the diagnostic tool: DataLogger - 4X4 Control Module

- While driving the vehicle at 48 km/h (30 mph), monitor the following wheel speed sensor PIDs:
  - Left Front Wheel Speed Sensor (LF_WSPD)
  - Left Rear Wheel Speed Sensor (LR_WSPD)
  - Right Front Wheel Speed Sensor (RF_WSPD)
  - Right Rear Wheel Speed Sensor (RR_WSPD)
- **Are all 4 wheel speeds within 3 km/h (1.8 mph) of each other?**

  **YES** : Go to F4.

  **NO** : Go to **Pinpoint Test G** .

**F4 CHECK FOR VALID ACCELERATOR PEDAL POSITION MONITORING ACCELERATOR PEDAL POSITION (AP) PID**

- Key ON, engine OFF (KOEO).
- Enter the following diagnostic mode on the diagnostic tool: DataLogger - 4X4 Control Module
- Monitor AP PID while pressing/releasing the accelerator pedal.
- **Does the accelerator position match the PID percent value?**

  **YES** : Go to F5.

  **NO** : REFER to the **INTRODUCTION - GASOLINE ENGINES** article to diagnose the engine control system.

**F5 CHECK FOR CORRECT 4X4 CONTROL MODULE OPERATION**

- Disconnect the 4X4 control module.
- Check the harness and component side connectors for:
  - corrosion.
  - pushed-out/bent pins.
- Check the ATCC solenoid jumper harness between the 4X4 control module and the ATCC solenoid, located on the rear subframe.
- Connect the 4X4 control module and make sure it seats correctly.
- Operate the system and determine if the concern is still present.
- **Is the concern still present?**

  **YES** : INSTALL a new 4X4 control module. REFER to **4X4 CONTROL MODULE**. CLEAR the DTCs. REPEAT the self-test.

  **NO** : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. CLEAR the DTCs. REPEAT the self-test.

**Pinpoint Test G: Tire/Axle Out of Acceptable Operating Range**

Refer to appropriate SYSTEM WIRING DIAGRAMS article for All Wheel Drive (AWD) schematic and connector information.

**Normal Operation**

The all wheel drive (AWD) system continuously monitors vehicle conditions and automatically adjusts the torque distribution between the front and rear wheels. During normal operation, most of the torque is sent to the front wheels. If wheel slip between the front and rear wheels is detected or the vehicle is under heavy acceleration, the AWD system increases torque to the rear wheels.

- DTC P1635 Tire/Axle Out of Acceptable Range - When the 4X4 control module detects inappropriate mini spare or road wheels/tires (greater than 5% at one wheel) installed, this DTC is set.

**This pinpoint test is intended to diagnose the following:**

- Wheels/tires
- Wheel speed sensors
- ABS module
- 4X4 control module

**PINPOINT TEST G: TIRE/AXLE OUT OF ACCEPTABLE OPERATING RANGE**

**NOTE: This test must be carried out on a hard surface in an area without traffic.**

**NOTE: Carry out Inspection and Verification as outlined before carrying out this pinpoint test.**

**G1 CHECK THE RECENT TIRE USAGE**

- Check with customer about recent tire usage or installation.
- **Has a tire been installed on the vehicle recently that was not originally supplied with the vehicle?**

  **YES** : INFORM the customer to only use tires of the type supplied with the vehicle. CLEAR the DTCs. REPEAT the self-test.

  **NO** : Go to G2.

**G2 CHECK TIRE SIZE AND BRAND**

- Check the tire size and the brand of tire.
- **Are all 4 tires the same size and brand?**

  **YES** : Go to G3.

  **NO** : INSTALL tires that are the same size and brand. INFORM the customer to only use the same size tires and brand. CLEAR the DTCs. REPEAT the self-test.

**G3 CHECK TIRE AIR PRESSURES**

- Check the air pressure in all 4 tires.
- **Are all 4 tires at the recommended air pressure?**

  **YES** : Go to G4.

  **NO** : ADJUST the tire air pressures. CLEAR the DTCs. REPEAT the self-test. INFORM the customer to maintain the correct tire air pressure.

**G4 CHECK FOR CORRECT WHEEL SPEEDS**

- Connect the diagnostic tool.
- Enter the following diagnostic mode on the diagnostic tool: DataLogger - 4X4 Control Module
- While driving the vehicle at 48 km/h (30 mph), monitor the following wheel speed sensor PIDs:
  - Left Front Wheel Speed Sensor (LF_WSPD)
  - Left Rear Wheel Speed Sensor (LR_WSPD)
  - Right Front Wheel Speed Sensor (RF_WSPD)
  - Right Rear Wheel Speed Sensor (RR_WSPD)
- **Are all 4 wheel speeds within 3 km/h (1.8 mph) of each other?**

  **YES** : Go to G5.

  **NO** : The ABS module is sending invalid wheel speed data to the 4X4 control module. REFER to **VEHICLE DYNAMIC SYSTEMS** article for additional ABS module diagnosis.

**G5 CHECK FOR CORRECT 4X4 CONTROL MODULE OPERATION**

- Disconnect: 4X4 Control Module C281
- Check for:
  - corrosion.
  - pushed-out pins.
- Connect the 4X4 control module and make sure it seats correctly.
- Operate the system and determine if the concern is still present.
- **Is the concern still present?**

  **YES** : INSTALL a new 4X4 control module. REFER to **4X4 CONTROL MODULE**. CLEAR the DTCs. REPEAT the self-test.

  **NO** : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. CLEAR the DTCs. REPEAT the self-test.

## REMOVAL AND INSTALLATION

### 4X4 CONTROL MODULE

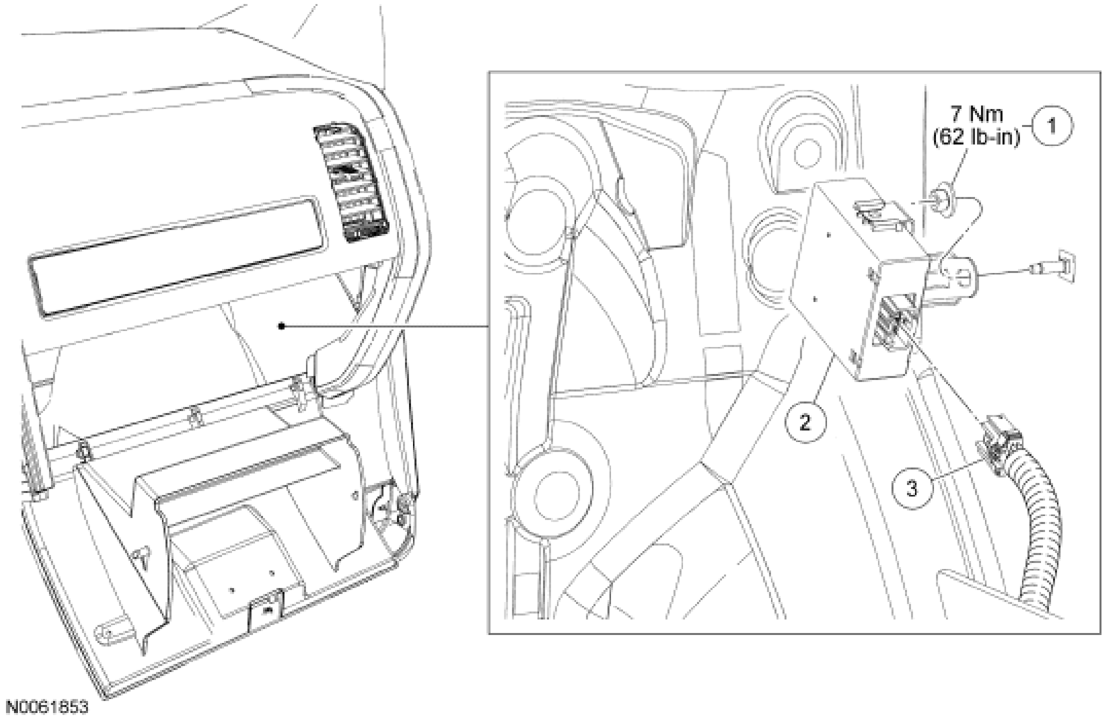


**Fig. 17: Exploded View Of 4X4 Control Module With Torque Specification**
**Courtesy of FORD MOTOR CO.**

| Item | Part Number | Description |
|---|---|---|
| 1 | N621906-S424 | 4X4 control module nut |
| 2 | 7E453 | 4X4 control module and bracket |
| 3 | - | Electrical connector (part of 14A005) |

**REMOVAL AND INSTALLATION**

1. From below the RH side instrument panel, remove the hush panel.
2. Disconnect the 4X4 control module harness connector.
3. Remove the 4X4 control module nut.
    - To install, tighten to 7 N.m (62 lb-in).
4. Remove the 4X4 control module and bracket.
5. To install, reverse the removal procedure.